

MACOMB, IL, USA

WMO: 722157

Lat: 40.520N

Lon: 90.652W

Elev: 216

StdP: 98.76

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04949

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.2	-16.2	-22.5	0.5	-18.4	-19.9	0.7	-15.4	13.8	-4.4	12.4	-4.5	3.3	300	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	33.0	24.2	32.1	24.0	30.8	23.3	26.5	31.0	25.5	29.9	24.6	28.7	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.8	29.3	24.1	19.5	28.0	23.0	18.3	27.0	84.0	31.0	79.3	30.0	75.4	28.7	29.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.7	10.2	8.8	-23.1	35.7	3.8	2.0	-25.9	37.1	-28.1	38.2	-30.2	39.3	-33.0	40.8		
			-23.4	28.1	3.7	0.7	-26.1	28.6	-28.2	29.1	-30.3	29.5	-33.0	30.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-3.6	-2.2	5.0	11.5	17.8	22.4	23.4	22.4	19.3	12.5	5.2	-1.0	(6)
(7)		DBStd	11.05	6.96	6.86	6.54	5.39	4.91	3.21	3.16	3.12	4.23	5.44	5.69	6.04	(7)
(8)		HDD10.0	1538	421	346	183	44	3	0	0	0	35	163	342		(8)
(9)		HDD18.3	3192	679	576	415	213	72	4	2	4	38	195	394	599	(9)
(10)		CDD10.0	1950	1	3	29	88	246	372	416	384	279	112	19	1	(10)
(11)		CDD18.3	564	0	0	2	7	56	127	160	131	67	14	0	0	(11)
(12)		CDH23.3	5429	0	0	21	104	532	1180	1581	1182	677	151	2	0	(12)
(13)		CDH26.7	1882	0	0	2	19	161	423	593	397	246	42	0	0	(13)
(14)	Wind	WSAvg	3.8	4.7	4.7	4.7	5.0	4.2	3.3	2.3	2.1	2.5	3.6	4.4	4.4	(14)
(15)	Precipitation	PrecAvg	957	47	46	63	94	124	109	106	96	81	71	71	47	(15)
(16)		PrecMax	1308	83	123	127	160	252	232	288	194	150	167	152	96	(16)
(17)		PrecMin	681	12	9	19	42	21	32	43	22	18	26	7	11	(17)
(18)		PrecStd	158	22	30	34	29	66	55	62	46	41	43	38	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.0	19.0	26.3	29.1	32.9	34.0	35.3	34.1	31.1	22.6	16.2		(19)
(20)		MCWB	10.6	13.2	18.2	18.3	21.7	24.4	25.8	24.4	21.7	21.4	13.8	14.4		(20)
(21)		2%	DB	11.3	14.3	22.2	26.5	30.2	32.4	32.9	32.4	31.8	27.3	18.9	12.7	(21)
(22)		MCWB	9.0	10.1	16.2	16.2	20.5	23.7	25.8	24.6	22.1	18.2	13.8	10.4		(22)
(23)		5%	DB	7.9	10.5	18.6	23.8	28.4	31.1	31.5	30.8	29.5	24.1	16.9	10.0	(23)
(24)		MCWB	6.1	7.6	14.0	15.6	19.5	23.2	24.9	24.1	21.8	17.3	12.7	8.3		(24)
(25)		10%	DB	5.0	7.1	15.2	21.3	26.7	29.1	30.0	28.8	27.5	21.6	14.0	7.3	(25)
(26)		MCWB	3.5	4.8	11.3	14.1	18.4	22.0	23.9	23.1	19.8	16.2	10.3	5.7		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.8	14.6	19.1	19.8	23.6	26.7	27.9	26.9	25.2	22.5	16.9	15.0	(27)
(28)		MCDWB	14.3	18.0	24.6	25.8	29.5	31.4	32.5	31.5	30.0	27.8	19.1	16.0		(28)
(29)		2%	WB	9.2	10.9	16.9	18.1	22.3	25.3	26.8	25.9	23.9	20.9	14.9	11.0	(29)
(30)		MCDWB	10.6	13.8	20.6	23.7	28.0	30.4	31.3	30.4	28.6	25.1	17.7	11.8		(30)
(31)		5%	WB	6.0	7.6	14.8	16.9	21.0	24.4	25.8	25.0	22.9	19.2	13.0	8.1	(31)
(32)		MCDWB	8.1	10.2	18.4	21.8	26.1	29.1	30.1	29.0	27.4	22.0	16.1	9.5		(32)
(33)		10%	WB	3.4	4.7	11.8	15.2	19.9	23.3	24.9	23.9	21.8	17.2	10.9	5.7	(33)
(34)		MCDWB	5.0	6.8	14.6	20.1	24.7	27.7	28.9	27.5	25.6	20.6	13.8	7.3		(34)
(35)	Mean Daily Temperature Range	MDBR	8.7	9.4	10.4	12.2	11.6	10.9	10.9	11.3	13.2	12.0	10.4	8.2		(35)
(36)		5% DB	MCDBR	11.7	14.7	13.9	16.1	13.7	12.6	11.7	12.7	14.3	15.1	14.2	11.5	(36)
(37)		MCWBR	9.4	10.5	8.3	8.1	5.8	5.9	5.7	6.3	6.1	7.3	9.2	9.5		(37)
(38)		5% WB	MCDBR	11.3	14.2	13.1	13.0	11.8	11.0	10.6	10.6	11.8	11.5	12.5	10.9	(38)
(39)		MCWBR	9.5	10.5	8.5	7.9	5.8	5.9	6.0	5.9	5.6	7.4	9.3	9.7		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.291	0.312	0.340	0.386	0.420	0.428	0.444	0.432	0.395	0.353	0.304	0.291		(40)
(41)		taud	2.456	2.386	2.402	2.272	2.241	2.271	2.199	2.244	2.310	2.427	2.509	2.467		(41)
(42)		Ebn at Noon	876	902	913	886	860	851	832	832	841	844	855	845		(42)
(43)		Edh at Noon	78	96	106	130	137	133	142	132	116	92	73	71		(43)
(44)	All-Sky Solar Radiation	RadAvg	1.82	2.78	3.73	4.76	5.54	6.22	6.26	5.48	4.71	3.16	2.11	1.55		(44)
(45)		RadStd	0.21	0.25	0.24	0.39	0.40	0.54	0.36	0.33	0.34	0.35	0.17	0.21		(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (35 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+60

Nomenclature: See separate page